# Matrices

Questions 1–122

**2.67 — Question 1.** It is given that $\mathbf{A} = \begin{pmatrix} 4 & 40 \\ 12 & 600 \end{pmatrix}$

a) Find $|\mathbf{A}|$

b) Find $10\begin{vmatrix} 4 & 4 \\ 12 & 60 \end{vmatrix}$

c) Find $40\begin{vmatrix} 1 & 1 \\ 12 & 60 \end{vmatrix}$

d) Find $480\begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix}$

**2.67 — Question 2.** It is given that $\mathbf{A} = \begin{pmatrix} 120 & 21 \\ 150 & 35 \end{pmatrix}$

a) Find $|\mathbf{A}|$

b) Find $2\begin{vmatrix} 60 & 21 \\ 75 & 35 \end{vmatrix}$

c) Find $6\begin{vmatrix} 20 & 21 \\ 25 & 35 \end{vmatrix}$

d) Find $30\begin{vmatrix} 4 & 21 \\ 5 & 35 \end{vmatrix}$

e) Find $150\begin{vmatrix} 4 & 21 \\ 1 & 7 \end{vmatrix}$

f) Find $1050\begin{vmatrix} 4 & 3 \\ 1 & 1 \end{vmatrix}$

g) Hence write 1050 as a product of its prime factors.

**2.67 — Question 3.** It is given that $\mathbf{A} = \begin{pmatrix} k & k+1 \\ 2k & 4(k+1) \end{pmatrix}$ where $k$ is a constant.

a) Find $|\mathbf{A}|$

b) Find $k\begin{vmatrix} 1 & k+1 \\ 2 & 4(k+1) \end{vmatrix}$

c) Find $2k\begin{vmatrix} 1 & k+1 \\ 1 & 2(k+1) \end{vmatrix}$

d) Find $2k(k+1)\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix}$

e) Hence write $|\mathbf{A}|$ in fully factorised form.

**2.67 — Question 4.** It is given that $\mathbf{A} = \begin{pmatrix} 18k-9 & 3-k \\ 6k-3 & 2k-6 \end{pmatrix}$ where $k$ is a constant.

a) Find $|\mathbf{A}|$

b) Find $(2k-1)\begin{vmatrix} 9 & 3-k \\ 3 & 2k-6 \end{vmatrix}$

c) Find $3(2k-1)\begin{vmatrix} 3 & 3-k \\ 1 & 2k-6 \end{vmatrix}$

d) Find $3(2k-1)(3-k)\begin{vmatrix} 3 & 1 \\ 1 & -2 \end{vmatrix}$

e) Hence write $|\mathbf{A}|$ in fully factorised form.

**2.67 — Question 5.** It is given that $\mathbf{A} = \begin{pmatrix} 4k & 6k-2 \\ 20k & 9k-3 \end{pmatrix}$, where $k$ is a constant. Completely factorise $|\mathbf{A}|$ without evaluating a determinant unless it is completely independent of $k$

**2.67 — Question 6.** It is given that $\mathbf{A} = \begin{pmatrix} 4-8k & 5-10k \\ 3k-6 & 4k-8 \end{pmatrix}$ where $k$ is a constant. By factorising $|\mathbf{A}|$, solve the equation $|\mathbf{A}| = 0$

**2.67 — Question 7.** It is given that $\mathbf{A} = \begin{pmatrix} 10 & 8 \\ 8 & 8 \end{pmatrix}$

a) Find $|\mathbf{A}|$

b) Let R1 and R2 be the first and second rows of $|\mathbf{A}|$ respectively. Perform the following steps in order:

- Factor 2 out of R1
- Factor 8 out of R2
- Let the new R1 = R1 − 4R2

c) Use your answer to b) to find $|\mathbf{A}|$

**2.67 — Question 8.** It is given that $\mathbf{A} = \begin{pmatrix} 16 & 20 \\ 9 & 5 \end{pmatrix}$

a) Find $|\mathbf{A}|$

b) Let C1 and C2 be the first and second columns of $|\mathbf{A}|$ respectively. Perform the following steps in order:

- Factor 5 out of C2
- Let the new C1 = C1 − 4C2

c) Use your answer to b) to find $|\mathbf{A}|$

**2.67 — Question 9.** It is given that $\mathbf{A} = \begin{pmatrix} 40 & 35 \\ 55 & 25 \end{pmatrix}$

a) Find $|\mathbf{A}|$

b) Let R1 and R2 be the first and second rows of $|\mathbf{A}|$ respectively, and C1 and C2 be the first and second columns of $|\mathbf{A}|$ respectively. Perform the following steps in order:

- Let the new C1 = C1 − 2C2
- Let the new R1 = R1 + 6R2

c) Use your answer to b) to find $|\mathbf{A}|$

**2.67 — Question 10.** It is given that $\mathbf{A} = \begin{pmatrix} 4k+5 & 53 \\ k & 12 \end{pmatrix}$ where $k$ is a constant.

a) Find $|\mathbf{A}|$

b) Let R1 and R2 be the first and second rows of $|\mathbf{A}|$ respectively, and C1 and C2 be the first and second columns of $|\mathbf{A}|$ respectively. Perform the following steps in order:

- Let the new R1 = R1 − 4R2
- Let the new C2 = C2 − C1

c) Use your answer to b) to find $|\mathbf{A}|$

**2.67 — Question 11.** Grace is attempting to factorise the determinant of the matrix $\mathbf{A} = \begin{pmatrix} q+1 & -5p \\ 3q+3 & 15p \end{pmatrix}$ where $p$ and $q$ are constants.
Her working and annotations are shown below:

| | |
|---|---|
| | $\begin{vmatrix} q+1 & -5p \\ 3q+3 & 15p \end{vmatrix}$ |
| (1) | $= \begin{vmatrix} q+1 & -5p \\ 3(q+1) & 15p \end{vmatrix}$ |
| (2) Factor $(q+1)$ out of C1 | $= (q+1)\begin{vmatrix} 1 & -5p \\ 3 & 15p \end{vmatrix}$ |
| (3) Factor 3 out of C1 | $= (q+1)\begin{vmatrix} 1 & -5p \\ 1 & 15p \end{vmatrix}$ |
| (4) Factor $5p$ out of C2 | $= 5p(q+1)\begin{vmatrix} 1 & -1 \\ 1 & 3 \end{vmatrix}$ |
| (5) Let the new C1 = C1 + C2: | $= 5p(q+1)\begin{vmatrix} 0 & -1 \\ 4 & 3 \end{vmatrix}$ |
| (6) Then: | $= 20p(q+1)$ |

a) Identify where Grace has made a mistake and explain what she has done wrong.
b) Write out fully correct workings for this question.

**2.67 — Question 12.** Fully factorise each of these determinants. Your aim in each case before evaluating the determinant is to have at least one element being 0

a)
$$\begin{vmatrix} x & y \\ x^2 & y^2 \end{vmatrix}$$

b)
$$\begin{vmatrix} x^2 & y^2 \\ xy & xy \end{vmatrix}$$

c)
$$\begin{vmatrix} xy & xy^2 \\ x^2 & xy \end{vmatrix}$$

d)
$$\begin{vmatrix} yz & xy \\ xyz & xyz \end{vmatrix}$$

e)
$$\begin{vmatrix} xyz & yz \\ xy & xyz \end{vmatrix}$$

f)
$$\begin{vmatrix} x^2y & xz^2 \\ xy^2 & xyz^2 \end{vmatrix}$$

g)
$$\begin{vmatrix} x^2y^2z & xy^2z^2 \\ x^2yz^2 & xyz \end{vmatrix}$$

**2.67 — Question 13.** Fully factorise each of these determinants, where $p$ and $q$ are constants. Your aim in each case before evaluating the determinant is to have at least one element being 0

a)

$$\begin{vmatrix} p & -p-2q \\ 4p+5q & p+2q \end{vmatrix}$$

b)

$$\begin{vmatrix} 2q-2p & 3p-3q \\ 14q-6p & 2p \end{vmatrix}$$

c)

$$\begin{vmatrix} 16p+5q-13 & 4-8p \\ 6q-12p & 6p-3 \end{vmatrix}$$

d)

$$\begin{vmatrix} 3-2q & 4q-8p \\ 6-4q & 8q+24p-30 \end{vmatrix}$$

**2.67 — Question 14.** Given that

$$|\mathbf{A}| = xy \begin{vmatrix} 1 & -3 \\ 2 & 4 \end{vmatrix}$$

show clearly, by using row and column operations, that

$$|\mathbf{A}| = \begin{vmatrix} x & -3y \\ 0 & 10y \end{vmatrix}$$

**2.67 — Question 15.** Given that

$$|\mathbf{A}| = x(x-1)y\begin{vmatrix} 2 & -3 \\ 3 & -1 \end{vmatrix}$$

show clearly, by using row and column operations, that

$$|\mathbf{A}| = \begin{vmatrix} 0 & 7y - 7xy \\ x & x + 2y - 2xy \end{vmatrix}$$

**2.67 — Question 16.** It is given that $\mathbf{A}$ is a singular matrix such that

$$\det \mathbf{A} = \begin{vmatrix} x - 5a & -4a \\ 14x + 2a & 9x + 7a \end{vmatrix}$$

where $a$ is a constant. Find the possible values of $x$ in terms of $a$

**2.67 — Question 17.** It is given that $\mathbf{A}$ is a singular matrix such that

$$\det \mathbf{A} = \begin{vmatrix} 8a^2 - 2x^2 & 3x^2 - 6ax - 24a^2 \\ 5ax - 10a^2 & 8ax - 32a^2 \end{vmatrix}$$

where $a$ is a constant such that $a \neq 0$.
Find the possible values of $x$ in terms of $a$

**2.67 — Question 18.** For constants $a$, $b$ and $c$, fully factorise

$$\begin{vmatrix} 3a - 4b - 3c & a - 2b - 2c \\ -a + 4b - c & a + 2b - 2c \end{vmatrix}$$

**2.67 — Question 19.** For constants $p$ and $q$, fully factorise

$$\begin{vmatrix} pq - 4p + 5q - 20 & 3pq + 15q \\ 3pq - 12p - 6q + 24 & 4pq - 10p - 8q + 20 \end{vmatrix}$$

**2.67 — Question 20.**

a) For constants $a$ and $b$, fully factorise

$$\begin{vmatrix} x + 3a + b & x + 3a \\ 5a + b & 5a \end{vmatrix}$$

b) It is given that the transformation $T$ can be represented by the matrix

$$A = \begin{pmatrix} x + 35 & x + 30 \\ 55 & 50 \end{pmatrix}$$

When a 2D shape of area 20 cm$^2$ is transformed by $T$, its new area is 50 cm$^2$.
Use your answer to part a) to find the possible values of $x$

**2.68 — Question 1.** Show that each of the following vectors are eigenvectors of the given matrix $\mathbf{A}$ and in each case state the associated eigenvalue:

a) $\begin{pmatrix} 2 \\ -1 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} 4 & 2 \\ 1 & 5 \end{pmatrix}$

b) $\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$

c) $\begin{pmatrix} 1 \\ -2 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix}$

**2.68 — Question 2.** It is given that $\mathbf{M} = \begin{pmatrix} 3 & -5 \\ 0 & -2 \end{pmatrix}$

a) Show that the characteristic equation of $\mathbf{M}$ is given by $\lambda^2 - \lambda - 6 = 0$ where $\lambda$ is an eigenvalue of $\mathbf{M}$

b) Hence state the eigenvalues of $\mathbf{M}$

**2.68 — Question 3.** Find the eigenvalues of each of the following matrices:

a) $\begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$

b) $\begin{pmatrix} -3 & 6 \\ -2 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 0 & -5 \\ 1 & 0 \end{pmatrix}$

d) $\begin{pmatrix} 2 & 7 \\ 0 & 2 \end{pmatrix}$

e) $\begin{pmatrix} -1 & 4 \\ -3 & 5 \end{pmatrix}$

f) $\begin{pmatrix} 6 & -1 \\ 2 & 3 \end{pmatrix}$

g) $\begin{pmatrix} \frac{1}{2} & -\frac{3}{4} \\ \frac{3}{4} & \frac{1}{2} \end{pmatrix}$

**2.68 — Question 4.** It is given that $\mathbf{M} = \begin{pmatrix} 5 & 2 \\ 1 & 4 \end{pmatrix}$

a) Show that the characteristic equation of $\mathbf{M}$ is given by $\lambda^2 - 9\lambda + 18 = 0$ where $\lambda$ is an eigenvalue of $\mathbf{M}$

b) Hence state the eigenvalues and corresponding eigenvectors of $\mathbf{M}$

**2.68 — Question 5.** Determine the eigenvalues and eigenvectors for each of the following:

a) $\begin{pmatrix} 4 & 2 \\ -3 & -3 \end{pmatrix}$

b) $\begin{pmatrix} -87 & 120 \\ -64 & 89 \end{pmatrix}$

c) $\begin{pmatrix} 60 & 10 \\ -10 & 40 \end{pmatrix}$

d) $\begin{pmatrix} 67 & -54 \\ 27 & -32 \end{pmatrix}$

e) $\begin{pmatrix} \frac{4}{15} & \frac{2}{3} \\ \frac{1}{3} & -\frac{1}{15} \end{pmatrix}$

f) $\begin{pmatrix} -\frac{26}{15} & \frac{4}{5} \\ \frac{4}{5} & \frac{2}{5} \end{pmatrix}$

**2.68 — Question 6.** Given that the matrix $\mathbf{M} = \begin{pmatrix} 6 & -2 \\ k-3 & 2 \end{pmatrix}$ has a repeated eigenvalue, find the value of $k$

**2.68 — Question 7.** Given that in each case, the matrix $\mathbf{M}$ has two distinct real eigenvalues, find the range of values for $k$ where $k$ is a real constant:

a) $\mathbf{M} = \begin{pmatrix} 4 & 1 \\ k & 3 \end{pmatrix}$

b) $\mathbf{M} = \begin{pmatrix} k & 2 \\ 3 & 5 \end{pmatrix}$

**2.68 — Question 8.** The matrix $\mathbf{M} = \begin{pmatrix} -1 & k \\ 4 & k-1 \end{pmatrix}$ where $k$ is a real constant.

a) Show that the characteristic equation of $\mathbf{M}$ is given by $\lambda^2 + (2-k)\lambda - (5k-1) = 0$ where $\lambda$ is an eigenvalue of $\mathbf{M}$

b) Find the range of values for $k$ for which $\mathbf{M}$ has two distinct real eigenvalues.

c) Given that $k = 9$, find the eigenvalues and corresponding eigenvectors of $\mathbf{M}$ giving your answer in terms of $k$

**2.68 — Question 9.** Show that the matrix

a) $\begin{pmatrix} p & -q \\ q & p \end{pmatrix}$ for constants $p$ and $q$ where $q \neq 0$ has complex eigenvalues $\lambda = p \pm qi$

b) $\begin{pmatrix} p & q \\ 0 & p \end{pmatrix}$ for constants $p$ and $q$ where $q \neq 0$ has eigenvalue given by $\lambda = p$

**2.68 — Question 10.** Prove that if $\lambda$ is an eigenvalue of the matrix $\mathbf{M}$, then $\lambda^n$ is an eigenvalue of $\mathbf{M}^n$ where $n$ is a positive whole number.

**2.68 — Question 11.** For each of the following, $\mathbf{v}$ is an eigenvector of both the matrix $\mathbf{A}$ and matrix $\mathbf{B}$. The corresponding eigenvalues for $\mathbf{v}$ in each case are also given, with an equation involving matrix $\mathbf{A}$, matrix $\mathbf{B}$, $\mathbf{v}$ and $\lambda$. Find the value of $\lambda$

a) $\lambda_A = 3, \lambda_B = 2, \mathbf{ABv} = \lambda\mathbf{v}$
b) $\lambda_A = 4, \lambda_B = 5, \mathbf{A}^2\mathbf{Bv} = \lambda\mathbf{v}$
c) $\lambda_A = -1, \lambda_B = 2, \mathbf{AB}^3\mathbf{v} = \lambda\mathbf{v}$
d) $\lambda_A = -2, \lambda_B = -4, \mathbf{A}^2\mathbf{B}^3\mathbf{v} = \lambda\mathbf{v}$
e) $\lambda_A = p, \lambda_B = q, \mathbf{A}^m\mathbf{B}^n\mathbf{v} = \lambda\mathbf{v}$ for constants $p$ and $q$ and positive constants $m$ and $n$

**2.69 — Question 1.** Matrix $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ has eigenvalues such that $\lambda_1 < \lambda_2$. Matrix $\mathbf{D}$ is diagonal and is defined such that $\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$.

a) By finding the eigenvectors of $\mathbf{A}$, find the matrix $\mathbf{P}$.
b) By finding $\mathbf{P}^{-1}\mathbf{AP}$, find the matrix $\mathbf{D}$.

**2.69 — Question 2.** For each of the following, find a possible matrix $\mathbf{P}$ and a possible diagonal matrix $\mathbf{D}$ such that $\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$:

a) $\mathbf{A} = \begin{pmatrix} 7 & 2 \\ 3 & 6 \end{pmatrix}$

b) $\mathbf{A} = \begin{pmatrix} 20 & 12 \\ 15 & 17 \end{pmatrix}$

c) $\mathbf{A} = \begin{pmatrix} 12 & 7 \\ -14 & -9 \end{pmatrix}$

**2.69 — Question 3.** Given the matrix $\mathbf{A} = \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$

a) Find the eigenvalues and corresponding eigenvectors of the matrix $\mathbf{A}$.

b) Explain why this matrix cannot be diagonalised.

**2.69 — Question 4.** Aaron is finding eigenvalues and eigenvectors to diagonalise the matrix $\mathbf{A} = \begin{pmatrix} 13 & 6 \\ 3 & 10 \end{pmatrix}$. His workings are shown below:

$$\lambda = 7,\ \lambda = 16,\ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ and } \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\mathbf{PDP}^{-1} = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 7 & 0 \\ 0 & 16 \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} -30 & 9 \\ 18 & -39 \end{pmatrix}$$

a) Find the two errors in his working.
b) Fully correct his working.

**2.69 — Question 5.** Find each of the following:

a) $\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}^8$

b) $\begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}^{10}$

c) $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}^k$ for constants $a$, $b$ and $k$

**2.69 — Question 6.** Show that if $\mathbf{M} = \mathbf{PDP}^{-1}$ then $\mathbf{M}^n = \mathbf{PD}^n\mathbf{P}^{-1}$.

**2.69 — Question 7.** Given the matrix $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$ and $\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$

a) Find the eigenvalues and corresponding eigenvectors of the matrix $\mathbf{A}$.
b) Find possible matrices $\mathbf{P}$ and $\mathbf{D}$.
c) Hence find $\mathbf{A}^4$.

**2.69 — Question 8.** Matrix $\mathbf{A} = \begin{pmatrix} -4 & 0 \\ -15 & 1 \end{pmatrix}$ has eigenvalues $\lambda_1 < \lambda_2$.

a) Find the eigenvalues of the matrix $\mathbf{A}$.
b) Find the corresponding eigenvectors for your eigenvalues.
c) Find possible matrices $\mathbf{P}$ and $\mathbf{D}$ such that $\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D}$.
d) Hence find $\mathbf{A}^{50}$, giving each element in terms of suitable powers of 2.

**2.69 — Question 9.** Given the matrix $\mathbf{M} = \frac{1}{4}\begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix}$ has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = \frac{1}{4}$ with corresponding eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

a) Find $\mathbf{M}^n$ by using $\mathbf{M}^n = \mathbf{PD}^n\mathbf{P}^{-1}$ where $\mathbf{D}$ is a diagonal matrix and $\mathbf{P}$ is the matrix of eigenvectors.
b) Given that $\mathbf{M}^n \to \mathbf{L}$ as $n \to \infty$, find the matrix $\mathbf{L}$.

**2.69 — Question 10.** Matrix $\mathbf{M} = \frac{1}{6}\begin{pmatrix} 4 & 2 \\ 1 & 5 \end{pmatrix}$ has eigenvalues $\lambda_1 < \lambda_2$.

a) Find $\mathbf{M}^n$ by using $\mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$ where $\mathbf{D}$ is a diagonal matrix and $\mathbf{P}$ is the matrix of eigenvectors.
b) Given that $\mathbf{M}^n \to \mathbf{L}$ as $n \to \infty$, find the matrix $\mathbf{L}$.

**2.69 — Question 11.** Given the matrix $\mathbf{A} = \begin{pmatrix} 4 & 3 \\ 0 & 2 \end{pmatrix}$. By considering $\mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$ where $\mathbf{D}$ is diagonal and $\mathbf{P}$ is the matrix of eigenvectors, show that

$$\mathbf{A}^n = \begin{pmatrix} 4^n & \frac{3}{2}(4^n - 2^n) \\ 0 & 2^n \end{pmatrix}.$$

**2.69 — Question 12.** Given the matrix $\mathbf{A} = \begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix}$. By considering $\mathbf{PD}^n\mathbf{P}^{-1}$ where $\mathbf{D}$ is diagonal and $\mathbf{P}$ is the matrix of eigenvectors, show that

$$\mathbf{A}^n = 5^{n-1}\begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix}.$$

**2.69 — Question 13.** Given the matrix $\mathbf{A} = \begin{pmatrix} p & q \\ p & q \end{pmatrix}$ where $p$ and $q$ are constants such that $p \neq -q$ and $q \neq 0$, show that

$$\mathbf{A}^n = (p+q)^{n-1}\mathbf{A}.$$

**2.69 — Question 14.** Given the matrix $\mathbf{A} = \begin{pmatrix} 1 & k+1 \\ k-1 & -1 \end{pmatrix}$ where $k$ is a constant and $k \neq \pm 1$ and $k \neq 0$, show that

$$\mathbf{A}^{2n+1} = k^{2n}\mathbf{A}.$$

**2.70 — Question 1.** A transformation $T$ is represented by the matrix $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$

a) State the geometrical transformation that $T$ represents.
b) Find the eigenvalues of matrix $\mathbf{A}$ and their corresponding eigenvectors.
c) State any invariant lines of the transformation.
d) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.

**2.70 — Question 2.** A transformation $T$ is represented by the matrix $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

a) State the geometrical transformation that $T$ represents.
b) Find the eigenvalues of matrix $\mathbf{A}$ and their corresponding eigenvectors.
c) State any invariant lines of the transformation.
d) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.

**2.70 — Question 3.** A transformation $T$ is represented by the matrix $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

a) State the geometrical transformation that $T$ represents.
b) Find the eigenvalues of $\mathbf{A}$
c) By considering the eigenvalues and then geometrically, explain why there are not any invariant lines of the transformation.
d) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.

**2.70 — Question 4.** A transformation $T$ is represented by the matrix $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$

a) Find the eigenvalues of matrix $\mathbf{A}$ and their corresponding eigenvectors.
b) State any invariant lines of the transformation.
c) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.
d) Describe fully the transformation represented by $\mathbf{A}$

**2.70 — Question 5.** A transformation $T$ is represented by the matrix $\mathbf{A} = \frac{1}{5}\begin{pmatrix} 13 & 6 \\ 6 & 22 \end{pmatrix}$

a) Find the eigenvalues of $\mathbf{A}$
b) Determine the Cartesian equations of the two lines passing through the origin which are invariant under the transformation $T$
c) Describe fully the transformation represented by $\mathbf{A}$

**2.70 — Question 6.** Under a linear transformation, $T$, the point $(3, 4)$ is mapped to the point $(6, 8)$
For the matrix representing $T$:

a) Write down the eigenvector.

b) Write down the corresponding eigenvalue.

**2.70 — Question 7.** Under a linear transformation, $T$, the point $(2, -5)$ is mapped to the point $(-4, 10)$
For the matrix representing $T$:

a) Write down the eigenvector.

b) Write down the corresponding eigenvalue.

**2.70 — Question 8.** A transformation $T$ is represented by the matrix $\mathbf{A} = \begin{pmatrix} p & q \\ q & p \end{pmatrix}$ for non-zero constants $p$ and $q$ where $p \neq q$

a) Find the eigenvalues and corresponding eigenvectors of matrix $\mathbf{A}$ giving your answer in terms of $p$ and $q$
b) Show that the invariant lines of the transformation are perpendicular.
c) Describe fully the transformation represented by $\mathbf{A}$

**2.70 — Question 9.** A transformation $T$ is represented by the matrix $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$

a) Find the eigenvalues of $\mathbf{A}$
b) Determine the Cartesian equations of the two lines passing through the origin which are invariant under the transformation $T$
c) Describe fully the transformation represented by $\mathbf{A}^2$

**2.70 — Question 10.** A transformation $T$ is represented by the matrix $\mathbf{A} = \begin{pmatrix} -10 & 8 \\ -16 & 14 \end{pmatrix}$

Describe fully the transformation represented by $\mathbf{A}^3$

**2.71 — Question 1.** It is given that $\mathbf{A} = \begin{pmatrix} 4 & 2 & 6 \\ 0 & 6 & 15 \\ 0 & 0 & 12 \end{pmatrix}$.

a) Find $|\mathbf{A}|$.

b) Find $2\begin{vmatrix} 2 & 1 & 3 \\ 0 & 6 & 15 \\ 0 & 0 & 12 \end{vmatrix}$.

c) Find $6\begin{vmatrix} 2 & 1 & 1 \\ 0 & 6 & 5 \\ 0 & 0 & 4 \end{vmatrix}$.

d) Find $12\begin{vmatrix} 1 & 1 & 1 \\ 0 & 6 & 5 \\ 0 & 0 & 4 \end{vmatrix}$.

e) Find $48\begin{vmatrix} 1 & 1 & 1 \\ 0 & 6 & 5 \\ 0 & 0 & 1 \end{vmatrix}$.

**2.71 — Question 2.** It is given that $\mathbf{A} = \begin{pmatrix} 22 & 0 & 0 \\ 660 & 15 & 0 \\ 220 & -12 & 2 \end{pmatrix}$.

a) Find $|\mathbf{A}|$.

b) Find $2\begin{vmatrix} 22 & 0 & 0 \\ 660 & 15 & 0 \\ 220 & -12 & 1 \end{vmatrix}$.

c) Find $4\begin{vmatrix} 11 & 0 & 0 \\ 330 & 15 & 0 \\ 110 & -12 & 1 \end{vmatrix}$.

d) Find $12\begin{vmatrix} 11 & 0 & 0 \\ 110 & 5 & 0 \\ 110 & -12 & 1 \end{vmatrix}$.

e) Find $60\begin{vmatrix} 11 & 0 & 0 \\ 22 & 1 & 0 \\ 110 & -12 & 1 \end{vmatrix}$.

f) Find $660\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 10 & -12 & 1 \end{vmatrix}$.

g) Hence write 660 as a product of its prime factors.

**2.71 — Question 3.** It is given that $\mathbf{A} = \begin{pmatrix} k & 3k^3 & 4k^3(k-1) \\ 0 & k^2 & 2k^2(k-1) \\ 0 & 0 & 3k^2(k-1) \end{pmatrix}$ where $k$ is a constant.

a) Find $|\mathbf{A}|$.

b) Find $k\begin{vmatrix} 1 & 3k^2 & 4k^2(k-1) \\ 0 & k^2 & 2k^2(k-1) \\ 0 & 0 & 3k^2(k-1) \end{vmatrix}$.

c) Find $k(k-1)\begin{vmatrix} 1 & 3k^2 & 4k^2 \\ 0 & k^2 & 2k^2 \\ 0 & 0 & 3k^2 \end{vmatrix}$.

d) Find $k^3(k-1)\begin{vmatrix} 1 & 3k^2 & 4 \\ 0 & k^2 & 2 \\ 0 & 0 & 3 \end{vmatrix}$.

e) Find $k^5(k-1)\begin{vmatrix} 1 & 3 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{vmatrix}$.

f) Hence write $|\mathbf{A}|$ in fully factorised form.

**2.71 — Question 4.** For constant $k$, it is given that

$$\mathbf{A} = \begin{pmatrix} 5k-10 & 0 & 0 \\ 4k-8 & 6-2k & 0 \\ 12-6k & 5k-15 & 3k-18 \end{pmatrix}.$$

a) Find $|\mathbf{A}|$.

b) Find $(k-2)\begin{vmatrix} 5 & 0 & 0 \\ 4 & 6-2k & 0 \\ -6 & 5k-15 & 3k-18 \end{vmatrix}$.

c) Find $(k-2)(k-3)\begin{vmatrix} 5 & 0 & 0 \\ 4 & -2 & 0 \\ -6 & 5 & 3k-18 \end{vmatrix}$.

d) Find $(k-2)(k-3)(k-6)\begin{vmatrix} 5 & 0 & 0 \\ 4 & -2 & 0 \\ -6 & 5 & 3 \end{vmatrix}$.

e) Hence write $|\mathbf{A}|$ in fully factorised form.

**2.71 — Question 5.** It is given that $\mathbf{A} = \begin{pmatrix} 24k & 4k+4 & 3k-6 \\ -6k & 2k+2 & 2k-4 \\ 12k & -3k-3 & 8-4k \end{pmatrix}$ where $k$ is a constant. Completely factorise $|\mathbf{A}|$ without evaluating a determinant unless it is completely independent of $k$.

**2.71 — Question 6.** It is given that

$$\mathbf{A} = \begin{pmatrix} 4k-2 & 8k-4 & 5-10k \\ 3-9k & 18k-6 & 6k-2 \\ -1-4k & 12k+3 & 20k+5 \end{pmatrix}$$

where $k$ is a constant. By factorising $|\mathbf{A}|$, solve the equation $|\mathbf{A}| = 0$.

**2.71 — Question 7.** It is given that $\mathbf{A} = \begin{pmatrix} 0 & -1 & 3 \\ 2 & 2 & 3 \\ 2 & 1 & -3 \end{pmatrix}$.

a) Find $|\mathbf{A}|$.

b) Let $R_1$, $R_2$ and $R_3$ be the first, second and third rows of $|\mathbf{A}|$ respectively. Perform the following steps in order:

- Let the new $R_1 = R_1 + R_3$.
- Let the new $R_2 = R_2 + R_3$.

c) Use your answer to b) to find $|\mathbf{A}|$.

**2.71 — Question 8.** It is given that $\mathbf{A} = \begin{pmatrix} 2 & 2 & -1 \\ 2 & 5 & 0 \\ 2 & 5 & 4 \end{pmatrix}$.

a) Find $|\mathbf{A}|$.

b) Let $R_1$, $R_2$ and $R_3$ be the first, second and third rows of $|\mathbf{A}|$ respectively, and let $C_1$ be the first column of $|\mathbf{A}|$. Perform the following steps in order:

- Factor 2 out of $C_1$.
- Let the new $R_3 = R_3 - R_2$.
- Let the new $R_2 = R_2 - R_1$.

c) Use your answer to b) to find $|\mathbf{A}|$.

**2.71 — Question 9.** By factoring out constants, and by using row and column operations, reduce the determinants below to the following format:

$$\begin{vmatrix} \# & \# & \# \\ 0 & \# & \# \\ 0 & 0 & \# \end{vmatrix}$$

and hence calculate the determinant.

a) $\begin{vmatrix} 1 & 0 & 5 \\ 2 & 1 & 2 \\ 3 & 2 & 1 \end{vmatrix}$

b) $\begin{vmatrix} 2 & 4 & 6 \\ 3 & -1 & 1 \\ -2 & 2 & 1 \end{vmatrix}$

c) $\begin{vmatrix} 4 & 1 & 6 \\ -1 & 3 & 3 \\ 4 & -2 & 3 \end{vmatrix}$

d) $\begin{vmatrix} 3 & 2 & 5 \\ -2 & 4 & 2 \\ -4 & -1 & 9 \end{vmatrix}$

**2.71 — Question 10.** It is given that $\mathbf{A} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -2 & 3 \\ 0 & 0 & 5 \end{pmatrix}$.

a) Find $|\mathbf{A}|$. Consider rotating the rows:

b) Find $\begin{vmatrix} 0 & 0 & 5 \\ 2 & 1 & 3 \\ 0 & -2 & 3 \end{vmatrix}$.

c) Find $\begin{vmatrix} 0 & -2 & 3 \\ 0 & 0 & 5 \\ 2 & 1 & 3 \end{vmatrix}$. Consider rotating the columns:

d) Find $\begin{vmatrix} 1 & 3 & 2 \\ -2 & 3 & 0 \\ 0 & 5 & 0 \end{vmatrix}$.

e) Find $\begin{vmatrix} 3 & 2 & 1 \\ 3 & 0 & -2 \\ 5 & 0 & 0 \end{vmatrix}$. Consider swapping two rows or two columns:

f) Find $\begin{vmatrix} 0 & -2 & 3 \\ 2 & 1 & 3 \\ 0 & 0 & 5 \end{vmatrix}$.

g) Find $\begin{vmatrix} 1 & 2 & 3 \\ -2 & 0 & 3 \\ 0 & 0 & 5 \end{vmatrix}$.

h) What do you notice?

**2.71 — Question 11.** For constants $a, b, c, d, e, f, g, h, i$, it is given that $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 6$. Write down the value of $\begin{vmatrix} g & h & i \\ d & e & f \\ a & b & c \end{vmatrix}$.

**2.71 — Question 12.** By reorganising the rows and/or columns only, rewrite the determinants below in the following format:

$$\begin{vmatrix} \# & \# & \# \\ 0 & \# & \# \\ 0 & 0 & \# \end{vmatrix}$$

and hence calculate the determinant.

a) $\begin{vmatrix} 3 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 5 & 0 \end{vmatrix}$

b) $\begin{vmatrix} 0 & 0 & 2 \\ 0 & 4 & 0 \\ -5 & 0 & 0 \end{vmatrix}$

c) $\begin{vmatrix} 0 & 0 & 6k \\ k & 0 & 0 \\ 0 & 4k & 0 \end{vmatrix}$, where $k$ is a constant.

d) $\begin{vmatrix} 0 & pq & 0 \\ p+q & 0 & 0 \\ 0 & 0 & pq \end{vmatrix}$, where $p$ and $q$ are constants.

**2.71 — Question 13.** Show clearly, without calculation, that each of the following determinants are equal to zero:

a) $\begin{vmatrix} 1 & 1 & 1 \\ 3 & 4 & -5 \\ 1 & 1 & 1 \end{vmatrix}$

b) $\begin{vmatrix} 2 & -3 & 4 \\ 3 & 0 & 6 \\ 4 & 2 & 8 \end{vmatrix}$

**2.71 — Question 14.** It is given that $\mathbf{A} = \begin{pmatrix} 4k+1 & k & 4k+1 \\ 3k & k & 3k \\ -2 & 4 & 2k+2 \end{pmatrix}$ where $k$ is a constant. Let $R_1$ and $R_2$ be the first and second rows of $|\mathbf{A}|$ respectively, and let $C_1$ and $C_3$ be the first and third columns of $|\mathbf{A}|$. Perform the following steps in order:

a) Let the new $R_1 = R_1 - R_2$; let the new $C_3 = C_3 - C_1$.

b) Use your answer to a) to find a fully factorised form for $|\mathbf{A}|$.

**2.71 — Question 15.** Show that each of these are independent of the constant $p$:

a) $\begin{vmatrix} 1 & p & -p \\ 1 & p+2 & 1 \\ -1 & -p & p+1 \end{vmatrix}$

b) $\begin{vmatrix} p-1 & 1 & p-1 \\ p-2 & 1 & p-2 \\ 3p-3 & p & 3p \end{vmatrix}$

c) $\begin{vmatrix} 2 & -p & -1 \\ 1 & 2p-5 & 2 \\ p & 2 & 1 \end{vmatrix}$

d) $\begin{vmatrix} p-2 & 3p-3 & p-1 \\ p+3 & p+14 & p+3 \\ p-1 & 3p-2 & p \end{vmatrix}$

**2.71 — Question 16.** Fully factorise each of these determinants, where $n$ is a constant:

a) $\begin{vmatrix} n+5 & 2n+4 & 9-n \\ 4-n & 4-n & 8-3n \\ n+3 & 2n+2 & 5 \end{vmatrix}$

b) $\begin{vmatrix} -2n & 1-3n & 1 \\ -4-n & 5-2n & 5 \\ -1-3n & 2-2n & n+2 \end{vmatrix}$

c) $\begin{vmatrix} 2n-4 & 3n-6 & 4-2n \\ -6 & 4n-6 & 2-5n \\ 3n-6 & 3n-6 & 2-n \end{vmatrix}$

d) $\begin{vmatrix} 9-15n & 12-17n & 12-15n \\ 40-49n & 40-49n & 40-50n \\ 13-20n & 16-22n & 16-20n \end{vmatrix}$

**2.71 — Question 17.** Riley is attempting to create his own question for a friend to revise from. He starts with $n^2(n+1)\begin{vmatrix} 1 & 2 & -3 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{vmatrix}$. His working and annotations are shown below:

1. Multiply $C_1$ by $(n+1)$, and multiply $R_2$ by $n$, and multiply $R_3$ by $n$:
$$= \begin{vmatrix} n+1 & 2 & -3 \\ 0 & n & 0 \\ 0 & 3n & n \end{vmatrix}.$$
2. Let the new $C_3 = C_3 + C_1$:
$$= \begin{vmatrix} n+1 & 2 & -3 \\ 0 & n & 0 \\ n+1 & 3n+2 & n-3 \end{vmatrix}.$$
3. Let the new $C_3 = C_3 + C_2$:
$$= \begin{vmatrix} n+1 & 2 & -1 \\ 0 & n & n \\ n+1 & 3n+2 & 4n+1 \end{vmatrix}.$$
4. Let the new $C_1 = C_1 - C_2$:
$$= \begin{vmatrix} n-1 & 2 & -1 \\ -n & n & n \\ -2n-1 & 3n+2 & 4n+1 \end{vmatrix}.$$

a) Identify where Riley has made a mistake.
b) Factorise the determinant in stage (4) of Riley's working as far as possible.

**2.71 — Question 18.** Fully factorise each of these determinants, where $n$ is a constant:

a) $\begin{vmatrix} n+1 & n^2 & 1 \\ -n-1 & n+1-n^2 & n \\ n+1 & 2n^2-n & 2n^2-n \end{vmatrix}$

b) $\begin{vmatrix} 3n-2 & n^2+2n & 2n-1 \\ 3n^2-1 & n+2 & 3n^2-1 \\ n^2+4n-4 & n^2+2n & n^2+3n-3 \end{vmatrix}$

c) $\begin{vmatrix} 3n-n^2 & n+1 & n^2-n-6 \\ 6+n-n^2 & -n^2-n-2 & n^2-9 \\ 6-2n & -n^2-3n-2 & n-3 \end{vmatrix}$

d) $\begin{vmatrix} n-7 & n-12 & n-10 \\ 5n^2-20n & 11n^2-44n & 10n^2-40n \\ 3n-6 & 5n-10 & 5n-8 \end{vmatrix}$

**2.71 — Question 19.** Jessica is attempting to factorise the determinant of the matrix

$$\mathbf{A} = \begin{pmatrix} 5p+q & -2q & 3p-3q \\ 9p & 9q & 6p-6q \\ 3pq+3q^2 & -3q^2 & pq-q^2 \end{pmatrix}$$

where $p$ and $q$ are constants. The beginnings of her working and annotations are shown below:

$$\begin{vmatrix} 5p+q & -2q & 3p-3q \\ 9p & 9q & 6p-6q \\ 3pq+3q^2 & -3q^2 & pq-q^2 \end{vmatrix}$$

(1) Factor $q$ out of $C_2$:

$$= q \begin{vmatrix} 5p+q & -2 & 3p-3q \\ 9p & 9 & 6p-6q \\ 3pq+3q^2 & -3q & pq-q^2 \end{vmatrix}$$

(2) Let the new $C_1 = C_1 - C_3$:

$$= q \begin{vmatrix} 2p+4q & -2 & 3p-3q \\ 3p+6q & 9 & 6p-6q \\ 2pq+4q^2 & -3q & pq-q^2 \end{vmatrix}.$$

Complete Jessica's working to write $|\mathbf{A}|$ in fully factorised form.

**2.71 — Question 20.** Fully factorise each of these determinants, where $p$, $q$ and $r$ are constants:

a) $\begin{vmatrix} 1 & p & pqr \\ 2 & q & pqr \\ 3 & r & pqr \end{vmatrix}$

b) $\begin{vmatrix} 1 & p & q+r \\ 2 & q & p+r \\ 3 & r & p+q \end{vmatrix}$

c) $\begin{vmatrix} 1 & p & pr \\ 2 & q & qr \\ 3 & r & pq \end{vmatrix}$

**2.71 — Question 21.** Fully factorise each of these determinants, where $p$, $q$ and $r$ are constants:

a) $\begin{vmatrix} p-q-r & p-q-r & p-q-r \\ p-q-r & 2p-q-r & p+3q-r \\ p-q-r & p-q & p-q \end{vmatrix}$

b) $\begin{vmatrix} -5q-7r & -3p-3q+3r & 2p-2q-4r \\ 3p-q-3r & p+q-r & 2p-2q-4r \\ p-6q-9r & -3p-3q+3r & 3p-3q-6r \end{vmatrix}$

c) $\begin{vmatrix} 2p+q+r & p-q & p+r \\ -2p+2r & -p+q-r & -2p-2r \\ 3p+q+2r & 2p+r & 2p+2r \end{vmatrix}$

d) $\begin{vmatrix} 2p-4q+r & 3p-3q & 3p-6q+3r \\ p-q & p-q & p-3q+2r \\ p-3q+r & 2p-2q & 2p-2q \end{vmatrix}$

**2.71 — Question 22.** Fully factorise each of these determinants, where $p$, $q$ and $r$ are constants:

a) $\begin{vmatrix} p & p^2 & p^3 \\ q & q^2 & q^3 \\ r & r^2 & r^3 \end{vmatrix}$

b) $\begin{vmatrix} 1 & p & p^3 \\ 1 & q & q^3 \\ 1 & r & r^3 \end{vmatrix}$

c) $\begin{vmatrix} 1 & p^2 & p^3 \\ 1 & q^2 & q^3 \\ 1 & r^2 & r^3 \end{vmatrix}$

d) $\begin{vmatrix} 1 & p & p^2(p+1) \\ 1 & q & q^2(q+1) \\ 1 & r & r^2(r+1) \end{vmatrix}$

**2.71 — Question 23.** By first showing that $(p+q+r)$ is a factor where $p$, $q$ and $r$ are constants, fully factorise each of these:

a) $\begin{vmatrix} 1 & p & q+r \\ 2 & q & p+r \\ 3 & r & p+q \end{vmatrix}$

b) $\begin{vmatrix} 1 & p+q & r \\ 2 & q+r & p \\ 4 & p+r & q \end{vmatrix}$

c) $\begin{vmatrix} 1 & 3 & 6 \\ p+2r & q+2p & r+2q \\ q-r & r-p & p-q \end{vmatrix}$

d) $\begin{vmatrix} p-2r & q-2p & r-2q \\ 2 & 4 & 8 \\ q+3r & r+3p & p+3q \end{vmatrix}$

**2.71 — Question 24.** Given that $|\mathbf{A}| = xy\begin{vmatrix}1 & 2 & 5\\0 & 1 & -3\\0 & 0 & 1\end{vmatrix}$, show clearly, by using row and column operations, that

$$|\mathbf{A}| = \begin{vmatrix}1+5y & 2x & 5y\\-3y & x & -3y\\4y & -x & 4y\end{vmatrix}.$$

**2.71 — Question 25.** Given that $|\mathbf{A}| = xy(x-1)(y-1)\begin{vmatrix}1 & 0 & 1\\-1 & 1 & -1\\0 & -1 & 1\end{vmatrix}$, show clearly, by using row and column operations, that

$$|\mathbf{A}| = \begin{vmatrix}xy-x & y & xy-x+y\\x-xy & x-y-1 & x-xy-y\\x-xy & 0 & x-xy\end{vmatrix}.$$

**2.71 — Question 26.** It is given that $\mathbf{A}$ is a singular matrix such that

$$\det \mathbf{A} = \begin{vmatrix} 6 & 3x+3a+5 & -5 \\ 5 & 4x+4a+4 & -4 \\ 4a-x+1 & 3a-2x+1 & x-4a-1 \end{vmatrix}$$

where $a$ is a constant. Find the possible values of $x$ in terms of $a$.

**2.71 — Question 27.** It is given that $\mathbf{A}$ is a singular matrix such that

$$\det \mathbf{A} = \begin{vmatrix} -2a & 6x+24a & -x-a \\ 2x-4a & 7x+28a & 3x-5a \\ 3x-7a & 13x+52a & 4x-8a \end{vmatrix}$$

where $a$ is a constant. Find the possible values of $x$ in terms of $a$.

**2.71 — Question 28.** It is given that $\mathbf{A}$ is a singular matrix such that $\det \mathbf{A} = \begin{vmatrix} 0 & 1 & x-p \\ x & p & px \\ p & q & p^2 \end{vmatrix}$ where $p$ and $q$ are constants.

a) Explain why $\det \mathbf{A} = \begin{vmatrix} 0 & 1 & x-p \\ x & p & px \\ p & q & 0 \end{vmatrix}$.

b) Hence express the possible values of $x$ in terms of $p$ and $q$.

**2.71 — Question 29.** Given $p$ and $q$ are constants, fully factorise

$$\begin{vmatrix} x^3 & 0 & p^2x \\ q & pq & q \\ x+q & p^2+pq & p+q \end{vmatrix}.$$

**2.71 — Question 30.** Given $p$ and $q$ are constants, fully factorise

$$\begin{vmatrix} q-x & -p^2 & x-q \\ x-q & px & q-p \\ x-q & p^2 & p+2q-x \end{vmatrix}.$$

**2.71 — Question 31.**

a) Given $a$ and $b$ are constants, fully factorise

$$\begin{vmatrix} a-2x+b & x+a & 4a-2x+b \\ a-2x+2b-1 & x+a+1 & 4a-2x+3b \\ 3a+b-x-1 & 1 & 6a+2b-2x \end{vmatrix}.$$

b) Transformation $T$ can be represented by

$$\mathbf{A} = \begin{pmatrix} 13-2x & x+6 & 31-2x \\ 19-2x & x+7 & 45-2x \\ 24-x & 1 & 50-2x \end{pmatrix}.$$

When a 3D shape of volume $300\,\text{cm}^3$ is transformed by $T$, its new volume is $1260\,\text{cm}^3$. Use your answer to part a) to find the possible values of $x$.

**2.72 — Question 1.** Show that each of the following vectors are eigenvectors of the given matrix $\mathbf{A}$ and in each case state the associated eigenvalue:

a) $\begin{pmatrix}1\\1\\0\end{pmatrix}$, $\mathbf{A} = \begin{pmatrix}2 & 1 & 0\\1 & 2 & 0\\0 & 0 & 4\end{pmatrix}$

b) $\begin{pmatrix}0\\1\\2\end{pmatrix}$, $\mathbf{A} = \begin{pmatrix}1 & -2 & 1\\1 & 1 & 1\\1 & 2 & 2\end{pmatrix}$

c) $\begin{pmatrix}1\\3\\2\end{pmatrix}$, $\mathbf{A} = \begin{pmatrix}-7 & 1 & 0\\-16 & 0 & 2\\-7 & -1 & 1\end{pmatrix}$

**2.72 — Question 2.** Given the matrix

$$\mathbf{M} = \begin{pmatrix}-11 & -6 & 10\\-4 & -9 & 8\\-10 & -12 & 14\end{pmatrix}$$

a) Show that the characteristic equation of $\mathbf{M}$ is given by $(\lambda - 3)(\lambda + 3)(\lambda + 6) = 0$ where $\lambda$ is an eigenvalue of $\mathbf{M}$

b) Hence state the eigenvalues of $\mathbf{M}$

**2.72 — Question 3.** Find the eigenvalues of each of the following matrices:

a)

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -3 \end{pmatrix}$$

b)

$$\begin{pmatrix} 1 & -2 & 0 \\ 2 & 1 & 5 \\ 0 & 0 & 5 \end{pmatrix}$$

c)

$$\begin{pmatrix} 0 & -1 & 1 \\ -1 & -1 & 2 \\ -1 & 0 & 1 \end{pmatrix}$$

d)

$$\begin{pmatrix} 1 & -1 & -3 \\ 0 & 4 & -8 \\ 0 & 4 & -4 \end{pmatrix}$$

**2.72 — Question 4.** Show that the matrix

$$\mathbf{A} = \begin{pmatrix} p & -q & 0 \\ q & p & 0 \\ 0 & 0 & r \end{pmatrix}$$

for constants $p$, $q$ and $r$ where $q \neq 0$ has complex eigenvalues $\lambda = p \pm q\mathrm{i}$ and $\lambda = r$

**2.72 — Question 5.** Given that the matrix

$$\mathbf{A} = \frac{1}{2}\begin{pmatrix} 4 & 0 & -3 \\ 0 & -10 & 11 \\ 0 & 0 & 1 \end{pmatrix}$$

a) Find the eigenvalues of $\mathbf{A}$
b) By first finding the matrix $\mathbf{A}^2$ calculate the eigenvalues of $\mathbf{A}^2$
c) Determine the eigenvalues of $\mathbf{A}^2$ in terms of the eigenvalues of $\mathbf{A}$
d) Show that in general, $\mathbf{M}^2\mathbf{v} = \lambda^2\mathbf{v}$ where $\lambda$ represents an eigenvalue of matrix $\mathbf{M}$

**2.72 — Question 6.** Given that the matrix

$$\mathbf{A} = \begin{pmatrix} 3 & -3 & 1 \\ 0 & -5 & 6 \\ 0 & -3 & 4 \end{pmatrix}$$

a) Find the eigenvalues of $\mathbf{A}$
b) By first finding the matrix $\mathbf{A}^3$ calculate the eigenvalues of $\mathbf{A}^3$
c) Determine the eigenvalues of $\mathbf{A}^3$ in terms of the eigenvalues of $\mathbf{A}$
d) Show that in general, $\mathbf{M}^3\mathbf{v} = \lambda^3\mathbf{v}$ where $\lambda$ represents an eigenvalue of matrix $\mathbf{M}$

**2.72 — Question 7.** Given the matrix

$$\mathbf{M} = \begin{pmatrix} -5 & 2 & 2 \\ -18 & 7 & 6 \\ -6 & 2 & 3 \end{pmatrix}$$

a) Show that $\mathbf{M}^2 + 3\mathbf{I} = k\mathbf{M}$ where $k$ is a constant to be found.
b) Hence show that

$$\mathbf{M}^{-1} = a\mathbf{M} + b\mathbf{I}$$

where $a$ and $b$ are constants to be found.
c) Find the eigenvalues of $\mathbf{M}$
d) Show that $\lambda^2 + 3 = k\lambda$ for each of your eigenvalues and your value of $k$

**2.72 — Question 8.** Given the matrix

$$\mathbf{M} = \begin{pmatrix} 1 & 3 & -1 \\ -2 & 3 & 2 \\ 1 & 0 & 2 \end{pmatrix}$$

a) Show that the characteristic equation of $\mathbf{M}$ is given by $(\lambda - 3)(\lambda^2 - 3\lambda + 9) = 0$ where $\lambda$ is an eigenvalue of $\mathbf{M}$
b) Hence find the eigenvalues and corresponding eigenvectors of $\mathbf{M}$

**2.72 — Question 9.** Find the eigenvectors for the following matrices for the given eigenvalue:

a) $\begin{pmatrix} 6 & 3 & 6 \\ -2 & -3 & -10 \\ -8 & -3 & -4 \end{pmatrix}$, $\lambda = 2$

b) $\begin{pmatrix} -2 & 5 & 7 \\ 1 & 0 & -1 \\ -1 & 1 & 2 \end{pmatrix}$, $\lambda = 0$

c) $\begin{pmatrix} 4 & -8 & 3 \\ 3 & -7 & 3 \\ 4 & -10 & 5 \end{pmatrix}$, $\lambda = -1$

d) $\begin{pmatrix} -1 & 1 & 2 \\ 4 & 2 & -1 \\ -4 & 1 & 3 \end{pmatrix}$, $\lambda = 3$

e) $\begin{pmatrix} -5 & 1 & 1 \\ \frac{3}{2} & 0 & 2 \\ -1 & 3 & 0 \end{pmatrix}$, $\lambda = -5$

**2.72 — Question 10.** Given the matrix

$$\mathbf{M} = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 1 & 1 & 5 \end{pmatrix}$$

a) Show that 5 is a repeated eigenvalue of $\mathbf{M}$

b) State all of the eigenvalues of $\mathbf{M}$

c) For each eigenvalue, find the corresponding eigenvectors.

**2.72 — Question 11.** Given the matrix

$$\mathbf{M} = \begin{pmatrix} 6 & 1 & -4 \\ 3 & 4 & -4 \\ 7 & 1 & -5 \end{pmatrix}$$

a) Show that 3 is an eigenvalue of $\mathbf{M}$
b) Find the other eigenvalues.
c) For each eigenvalue, find the corresponding eigenvectors.

**2.72 — Question 12.** Given the matrix

$$\mathbf{M} = \begin{pmatrix} -4 & 16 & -8 \\ -5 & 13 & -6 \\ -10 & 22 & -10 \end{pmatrix}$$

a) Show that 2 is an eigenvalue of $\mathbf{M}$
b) Find the other eigenvalues.
c) For each eigenvalue, find the corresponding eigenvectors.

**2.72 — Question 13.** Given the matrix

$$\mathbf{M} = \begin{pmatrix} -1 & p & 0 \\ p & 0 & 2 \\ 0 & 2 & 1 \end{pmatrix}$$

where $p$ is a real constant.
One of the eigenvalues of matrix $\mathbf{M}$ is 3
The corresponding eigenvector is $\begin{pmatrix} 1 \\ 2 \\ q \end{pmatrix}$ for constant $q$
Find the value of $p$ and $q$

**2.72 — Question 14.** It is given that matrix

$$\mathbf{M} = \frac{1}{14}\begin{pmatrix} 10 & -44 & -2k \\ 0 & 42 & 0 \\ 6 & -53 & -17 \end{pmatrix}$$

where $k$ is a positive integer.
It is known that the eigenvalue $\lambda_1 = 3$ has corresponding eigenvector $\mathbf{v}_1$
The other two eigenvectors are $\begin{pmatrix} a \\ 0 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$
where $a$ is a positive integer.

a) Find the value of $k$
b) Determine the value of $a$
c) State the values of all the eigenvalues.

**2.72 — Question 15.** Given the matrix

$$\mathbf{M} = \begin{pmatrix} 6 & -2 & 2 \\ 2-k & 2+k & k-2 \\ 4-k & k-4 & k+4 \end{pmatrix}$$

where $k$ is a real constant. Two of the eigenvalues of matrix $\mathbf{M}$ are 4 and 8

a) Find the third eigenvalue of $\mathbf{M}$ giving your answer in terms of $k$

The eigenvectors of $\mathbf{M}$ corresponding to the eigenvalues 4 and 8 are $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ respectively.

b) Find the eigenvector corresponding to the third eigenvalue.

**2.72 — Question 16.** Given the matrix

$$\mathbf{M} = \begin{pmatrix} k-2 & -3 & 3 \\ 6 & -5 & 3 \\ 0 & 0 & -2 \end{pmatrix}$$

where $k$ is an integer,

a) find the value of $k$ for which $\mathbf{M}$ has a repeated eigenvalue.

For that value of $k$,

b) find all the eigenvalues and corresponding eigenvector.

c) Find the trace($\mathbf{M}$)

**2.72 — Question 17.**

$$\mathbf{M} = \begin{pmatrix} 3a-b-c & -a+b & -a+c \\ 3a-2b-c & -a+2b & -a+c \\ 3a-b-2c & -a+b & -a+2c \end{pmatrix}$$

where $a$, $b$ and $c$ are constants.
Giving your answer in terms of $a$, $b$ and $c$:

a) Find $\det \mathbf{M}$
b) Find the eigenvalues of $\mathbf{M}$
c) Find the trace($\mathbf{M}$)

**2.72 — Question 18.** Given that the matrix

$$\mathbf{M} = \frac{1}{2}\begin{pmatrix} a+c & a-c & c-a \\ a-b & a+b & b-a \\ c-b & b-c & b+c \end{pmatrix}$$

where $a$, $b$ and $c$ are constants.
Giving your answer in terms of $a$, $b$ and $c$:

a) Find $\det \mathbf{M}$
b) Find the eigenvalues of $\mathbf{M}$
c) Find the trace($\mathbf{M}$)

**2.73 — Question 1.** It is given that $\mathbf{A} = \begin{pmatrix} 1 & 0 & -4 \\ 0 & 2 & -5 \\ 0 & 0 & -3 \end{pmatrix}$. Two of the eigenvalues of $\mathbf{A}$ are $\lambda_1 = -3$ and $\lambda_2 = 1$ with eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ respectively. A matrix $\mathbf{D}$ is diagonal and is defined such that $\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$.

a) Find the value of the third eigenvalue $\lambda_3$.

b) By finding the eigenvector corresponding to $\lambda_3$, determine a possible matrix $\mathbf{P}$ and hence $\mathbf{P}^{-1}$.

c) By finding $\mathbf{P}^{-1}\mathbf{AP}$, find a possible matrix $\mathbf{D}$.

**2.73 — Question 2.** For each of the following, by finding eigenvalues $\lambda_1 < \lambda_2 < \lambda_3$ and their respective eigenvectors to find a matrix $\mathbf{P}$, and using your calculator to find $\mathbf{P}^{-1}$, determine diagonal matrix $\mathbf{D}$ such that $\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$:

a) $\mathbf{A} = \begin{pmatrix} 5 & 1 & -5 \\ 2 & 3 & -4 \\ 2 & 1 & -2 \end{pmatrix}$

b) $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 4 & 3 \end{pmatrix}$

**2.73 — Question 3.** Given the matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$:

a) Find the eigenvalues and corresponding eigenvectors of matrix $\mathbf{A}$.

b) Explain why this matrix cannot be diagonalised.

**2.73 — Question 4.** Find each of the following:

a) $\begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^6$

b) $\begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{pmatrix}^9$

c) $\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}^k$ for constants $a, b, c$ and $k$.

**2.73 — Question 5.** The matrix $\mathbf{A}$ has eigenvalues $-1, 1, 3$ with corresponding eigenvectors $\begin{pmatrix}1\\0\\1\end{pmatrix}$, $\begin{pmatrix}2\\1\\-1\end{pmatrix}$ and $\begin{pmatrix}0\\1\\3\end{pmatrix}$ respectively. Determine the matrix $\mathbf{A}^4$.

**2.73 — Question 6.** The matrix $\mathbf{M}$ has eigenvalues $-5, -1, 1$ with corresponding eigenvectors $\begin{pmatrix}0\\2\\1\end{pmatrix}$, $\begin{pmatrix}1\\1\\1\end{pmatrix}$ and $\begin{pmatrix}1\\0\\3\end{pmatrix}$ respectively. Show that $\mathbf{M}^{10} = \begin{pmatrix} 1 & 0 & 0 \\ -6\times 5^9 + \frac{6}{5} & 4\times 5^9 + \frac{1}{5} & 2\times 5^9 - \frac{2}{5} \\ -3\times 5^9 + \frac{3}{5} & 2\times 5^9 - \frac{2}{5} & 5^9 + \frac{4}{5} \end{pmatrix}$.

**2.73 — Question 7.** Given that the matrix $\mathbf{M}$ has eigenvalues $\lambda_1 = \frac{1}{2}$, $\lambda_2 = \frac{1}{3}$ and $\lambda_3 = 1$ with corresponding eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and $\mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$:

a) Find $\mathbf{M}^n$ giving your answer in terms of $n$.

b) Given that $\mathbf{M}^n \to \mathbf{L}$ as $n \to \infty$, find the matrix $\mathbf{L}$.

**2.73 — Question 8.** Given that the matrix $\mathbf{M}$ has eigenvalues $\lambda_1 = 1, \lambda_2 = \frac{3}{4}$ and $\lambda_3 = \frac{1}{5}$ with eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$ and $\mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$:

a) Find $\mathbf{M}^n$ giving your answer in terms of $n$.

b) Given that $\mathbf{M}^n \to \mathbf{L}$ as $n \to \infty$, find the matrix $\mathbf{L}$.

**2.73 — Question 9.** Given the matrix $\mathbf{A} = \begin{pmatrix} k & 0 & k \\ 0 & k & 0 \\ k & 0 & k \end{pmatrix}$, where $k$ is a non-zero constant. Show that $\mathbf{A}^n = k^n \begin{pmatrix} 2^{n-1} & 0 & 2^{n-1} \\ 0 & 1 & 0 \\ 2^{n-1} & 0 & 2^{n-1} \end{pmatrix}$.

**2.73 — Question 10.** Given the matrix $\mathbf{A} = \begin{pmatrix} a - \frac{b}{2} & \frac{b}{2} & -a + \frac{b}{2} \\ a - b & b & -a + b \\ -\frac{b}{2} & \frac{b}{2} & \frac{b}{2} \end{pmatrix}$, where $a$ and $b$ are non-zero positive constants and $a < b$:

a) Find the eigenvalues of $\mathbf{A}$ where $\lambda_1 < \lambda_2 < \lambda_3$.

b) Hence, show that $\mathbf{A}^n = \frac{1}{2} \begin{pmatrix} 2a^n - b^n & b^n & b^n - 2a^n \\ 2a^n - 2b^n & 2b^n & 2b^n - 2a^n \\ -b^n & b^n & b^n \end{pmatrix}$.

**2.74 — Question 1.** Each of the following are an eigenvalue with its corresponding eigenvector for a matrix $\mathbf{M}$ In each case, describe the effect of the transformation in the direction of the given eigenvector:

a) $\lambda = 2, \mathbf{v} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

b) $\lambda = 4, \mathbf{v} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

c) $\lambda = 1, \mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

d) $\lambda = -1, \mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$

e) $\lambda = -2, \mathbf{v} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

**2.74 — Question 2.** A transformation $T$ is represented by the matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

a) Find the eigenvalues of matrix $\mathbf{A}$ and their corresponding eigenvectors.

b) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.

c) Describe fully the transformation represented by $\mathbf{A}$

**2.74 — Question 3.** A transformation $T$ is represented by the matrix $\mathbf{A} = \begin{pmatrix} -1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$

a) Find the eigenvalues of matrix $\mathbf{A}$ and their corresponding eigenvectors.
b) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.
c) Describe fully the transformation represented by $\mathbf{A}$

**2.74 — Question 4.** A transformation $T$ is represented by the matrix $\mathbf{A} = \begin{pmatrix} 3 & -4 & -2 \\ 0 & -1 & 0 \\ 0 & 2 & 1 \end{pmatrix}$

a) Find the eigenvalues of matrix $\mathbf{A}$ and their corresponding eigenvectors.
b) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.
c) Describe fully the transformation represented by $\mathbf{A}$

**2.74 — Question 5.** Under a linear transformation, $T$, the point $(3, 4, -1)$ is mapped to the point $(-15, -20, 5)$ For the matrix representing $T$:

a) Write down the eigenvector.

b) Write down the corresponding eigenvalue.

**2.74 — Question 6.** A transformation $T$ is represented by the matrix $\mathbf{A}$. Under the transformation, points on the line with direction $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ are stretched by scale factor 2. Points along the $x$-axis are invariant. Points along the $z$-axis are stretched by scale factor 3 and reflected across the origin.

a) Describe fully the transformation represented by $\mathbf{A}^n$ giving your answers in terms of $n$

b) Determine the matrix $\mathbf{A}$

c) Determine the matrix $\mathbf{A}^n$

**2.74 — Question 7.** A transformation $T$ is represented by the matrix $\mathbf{A} = \begin{pmatrix} p & q & 0 \\ q & p & 0 \\ 0 & 0 & r \end{pmatrix}$ for non-zero constants $p, q$ and $r$ where $p < q < r$

a) Find the eigenvalues and corresponding eigenvectors of matrix $\mathbf{A}$ giving your answer in terms of $p, q$ and $r$

b) Show that the three invariant lines are mutually perpendicular.

c) Describe fully the transformation represented by $\mathbf{A}$

**2.74 — Question 8.** A transformation $T$ is represented by the matrix

$$\mathbf{A} = \begin{pmatrix} 1-k & k & k \\ -2 & k+1 & 2 \\ 2-2k & k & 2k-1 \end{pmatrix}$$

where $k$ is a constant where $k > 2$

The eigenvalues of $\mathbf{A}$ are $\lambda_1 < \lambda_2 < \lambda_3$

a) Find the eigenvalues of $\mathbf{A}$

The eigenvectors of $\mathbf{A}$ are given by

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \text{ and } \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

b) Describe fully the transformation represented by $\mathbf{A}^3$ giving your answers in terms of $k$